



Cognivara

ACADEMY · BUILD LEARN ENGINEER

2026 EDITION · FOUNDING BATCH

Advanced **GenAI** Engineer Program

A live, mentor-led program for engineers who want to design, evaluate, deploy and defend production-grade GenAI systems — not just build demos.

16 Weeks

DURATION

Live Online

FORMAT

8–10 PM IST

TIMING

20 Max

COHORT SIZE

₹49,999

ONE-TIME FEE

ABOUT · THE PROGRAM

Learn technology the way it's **actually built**.

Cognivara Academy is a live, mentor-led learning academy focused on Generative AI, Data Science, Cloud and DevOps. We believe engineers learn best by building, breaking, debugging and shipping — not by watching a syllabus go by.

THE PROMISE

Stop building demos. Start building production systems.

You'll learn by building: a deployed GenAI API, an evaluated retrieval system, a fine-tuned model, and the architecture judgment to defend your design choices in a technical interview.

RAG

Hands-on vector databases, hybrid search and evaluation.

Agents

Built with LangGraph, MCP, guardrails and security.

Fine-tuning

LoRA, QLoRA, distillation and real serving workflows.

Production

Docker, AWS, CI/CD and cost-aware deployment.

Built around a simple learning cycle

Live instructor-led

Interactive online sessions, not prerecorded lectures.

Session recordings

Every live session is recorded for enrolled learners.

1:1 doubt support

Dedicated time to clear technical blockers.

Project reviews

Direct feedback on your implementation.

Career support

Resume, GitHub, mock interviews, job-search strategy.

Small cohorts

Capped at 20 engineers for real mentor attention.

Current curriculum

Tools and workflows reviewed as the industry changes.

Portfolio-ready

A capstone you can show in interviews and on GitHub.

HOW LEARNING WORKS

A loop that repeats through every module

1

Learn

Live session with your instructor

2

Practice

Guided hands-on labs

3

Build

Apply it in a real project

4

Review

Mentor feedback on your work

5

Improve

Refine based on feedback

6

Prepare

Interview-ready explanation

Built for engineers ready to go deeper

Prior GenAI experience helps — but it isn't required.

DESIGNED FOR

- Software engineers and backend / full-stack developers
- ML engineers and data scientists
- Professionals transitioning into GenAI roles
- Students and professionals with an IT background

NOT THE RIGHT FIT FOR

- Absolute beginners without a coding background
- Learners who want to avoid hands-on debugging
- People looking for a passive, video-only format
- Anyone expecting a guaranteed job outcome

PREREQUISITES

- Comfort with Python, basic SQL and Git/GitHub
- Familiarity with REST APIs and foundational ML concepts
- A laptop with a reliable internet connection

Six phases. One evolving system.

Each phase builds on the last. You ship real, working components incrementally instead of only learning concepts in isolation — by week 16 they come together into one capstone system.

1

WEEKS 1–3

Engineering Foundations

FastAPI, async Python, embeddings, chunking and transformer architecture.

2

WEEKS 4–7

Production RAG

Vector search, hybrid retrieval, reranking and automated RAG evaluation.

3

WEEKS 8–10

Agents, Tools & Security

LangGraph, tool calling, MCP, prompt-injection defense and guardrails.

4

WEEKS 11–13

Fine-Tuning Engineering

LoRA, QLoRA, evaluation against overfitting, distillation and serving.

5

WEEK 14

Deploy

Docker, CI/CD, AWS EC2/ECR, caching, cost tracking and model routing.

6

WEEKS 15–16

Integrate & Defend

Full system assembly, live demo, architecture defense and interview prep.

Engineering Foundations → Production RAG

Phase 1 · Weeks 1-3

Engineering Foundations

- **Week 1 — FastAPI & Async Python:** `async/await`, `Pydantic`, streaming responses, server-sent events and non-blocking LLM calls.
- **Week 2 — Embeddings & Chunking:** tokenization, semantic chunking, local LLMs and INT8/INT4 quantization.
- **Week 3 — Transformer Architecture:** self-attention, KV cache, time-to-first-token, speculative decoding and inference benchmarking.

Phase 2 · Weeks 4-7

Production RAG Systems

Build retrieval systems that are accurate, fast and measurable — not just functional.

- Vector search: FAISS to Qdrant, HNSW indexing, metadata filtering
- Hybrid search (BM25 + dense retrieval) with cross-encoder reranking
- Advanced strategies: RAG-Fusion, HyDE, multi-query, reciprocal rank fusion
- Evaluation and observability: RAGAS, LLM-as-judge, LangSmith, Phoenix tracing

Agents, Tools & Security → Fine-Tuning

Phase 3 · Weeks 8–10

Agents, Tools & Security

- **Week 8 — LangGraph & Stateful Agents:** graph-based workflows, conditional routing, cyclic graphs, memory and planning.
- **Week 9 — Tool Calling & MCP:** function schemas, MCP architecture, multi-agent coordination and human-in-the-loop review.
- **Week 10 — Security & Guardrails:** prompt-injection defense, input/output validation and least-privilege tool access.

Phase 4 · Weeks 11–13

Fine-Tuning Engineering

Weeks 11–13 cover instruction tuning, LoRA/QLoRA, dataset construction, evaluating against overfitting and catastrophic forgetting, and serving via distillation and quantization.

IF YOU NEED	USE THIS APPROACH
Current information or citations	RAG
A consistent output format	Prompting
A specific style or behaviour	Fine-tuning
Lower inference cost at scale	Fine-tuning
External actions	Agents / tools

Deploy, Integrate & Defend

The final three weeks bring every earlier phase together into one deployed, defensible system.

Week 14 — Deploy

Docker, CI/CD, AWS EC2/ECR, Redis caching, cost tracking and model routing.

Week 15 — Integrate

Full system assembly, end-to-end testing, architecture review, cost and performance analysis.

Week 16 — Defend

Live demo, architecture defense, interview preparation and a portfolio-ready deliverable.

Capstone principle: you deploy a fine-tuned model only if evaluation proves it is the better engineering choice over RAG or prompting — not by default.

CAPSTONE PROJECT

A real production system you'll deploy

Build one end-to-end GenAI system across a domain of your choice — Finance, Legal, Healthcare or Enterprise — designed to ship, measured with evidence, and strong enough for your portfolio.

- A fully deployed, production-ready GenAI application
- An evaluation report with metrics and trade-offs
- Cost, latency and quality analysis
- Architecture documentation you can present confidently
- A public GitHub project recruiters can review

How the capstone is evaluated

COMPONENT	WEIGHT
RAG quality & evaluation	20%
System architecture	15%
Agent / tool reliability	15%
Fine-tuning methodology	15%
Security	10%
Performance & cost	10%
Deployment	10%
Technical presentation	5%

TOOLS & INSTRUCTOR

What you'll work with

Python

FastAPI

LLM APIs

Embeddings

FAISS

Qdrant

RAG

LangGraph

MCP

RAGAS

LangSmith

LoRA

QLoRA

Docker

AWS

CI/CD

Redis

GitHub Actions



Roshan Salunke — Data Science Engineer

2+ years building production AI/ML and GenAI applications.

- Expertise in LLMs, agentic AI, RAG pipelines and scalable data systems
- Built a financial RAG assistant serving 1,660+ companies and 73,000+ reports
- Reduced inference latency from 3.0s to 1.2s and storage by 94%
- Trained 100+ engineers in data science and NLP with 90%+ satisfaction

SCHEDULE & INVESTMENT

16 weeks. About 160 live hours. One fee.

16	10	160	20
WEEKS	HRS / WEEK	TOTAL LIVE HRS	MAX COHORT

Fee	₹49,999 one-time
EMI	Not available
Time slot	8:00–10:00 PM IST, Mon–Fri
Cloud / GPU budget	₹2,000–5,000 (external, as used)
Certificate	Yes, on completing requirements
Placement guarantee	No

We provide career and placement assistance — resume review, GitHub/portfolio review, mock interviews and job-search strategy. This does not constitute a guarantee of employment, salary, company or role.

FAQ

Frequently asked questions

Q. Do I need prior GenAI experience?

No, but Python proficiency is required to keep pace with the labs.

Q. Will I fine-tune a real model?

Yes — you'll apply QLoRA to a small open-weight model during Phase 4.

Frequently asked questions

Q. What if I miss a live session?

Recordings are available for enrolled learners for every live session.

Q. Is there a certificate?

Yes, awarded on meeting the program's completion requirements.

Q. Do I need to pay for GPU or cloud usage?

Budget roughly ₹2,000–5,000 for GPU/AWS usage during the program.

Q. Are EMI plans available?

Not currently – the program fee is a one-time payment.

Q. Is placement guaranteed?

No. We provide placement assistance, not a guarantee of employment.

Q. Can I use the capstone in my portfolio?

Yes, it's designed as a public GitHub project you can show recruiters.

ENROLLMENT

Ready to build your next skill?

Learn with mentors. Build real projects. Prepare for your next opportunity.

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Placement assistance does not constitute a job guarantee. Program fees, schedules, curriculum and access terms are subject to the applicable enrollment terms shared before payment.
